Integrative Modeling of Doxorubicin Response in Cancer

İstanbul Bilgi University

Faculty of Engineering and Natural Sciences Department of Genetics and Bioengineering

Cansu Akdemir 122201043 Efehan Evren 121201060 Esra Rençberler 122201045

Supervised by Asst. Prof. Dr. Özlem Ulucan Açan

Aim

The aim of this study is to develop an integrative computational framework that links gene expression–based IC50 estimation with time-dependent modeling of apoptotic dynamics in doxorubicin-treated cancer cells. By combining static drug sensitivity prediction with dynamic apoptosis modeling, this framework seeks to provide a more mechanistic and temporally resolved characterization of cellular response to chemotherapy.

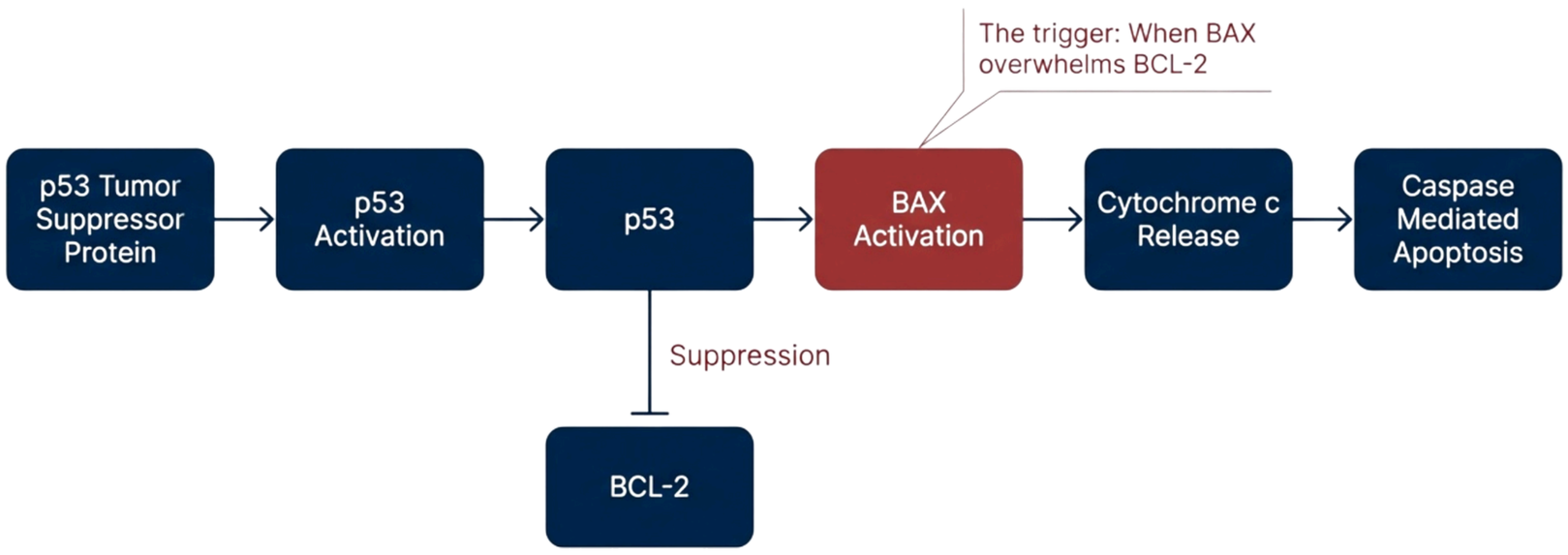
Motivation

In cancer research, drug sensitivity is commonly assessed using static metrics such as IC50. Although IC50 is widely used due to its simplicity and comparability across studies, it represents a single time-point measurement and fails to capture the temporal progression of drug-induced cell death. Apoptosis is a dynamic, multistep process involving initiation, regulation, and irreversible commitment. As a result, cancer cells with similar IC50 values may exhibit substantially different apoptotic behaviors over time. This limitation highlights the need for modeling approaches that go beyond static sensitivity measures and incorporate time-dependent molecular mechanisms underlying chemotherapy response.

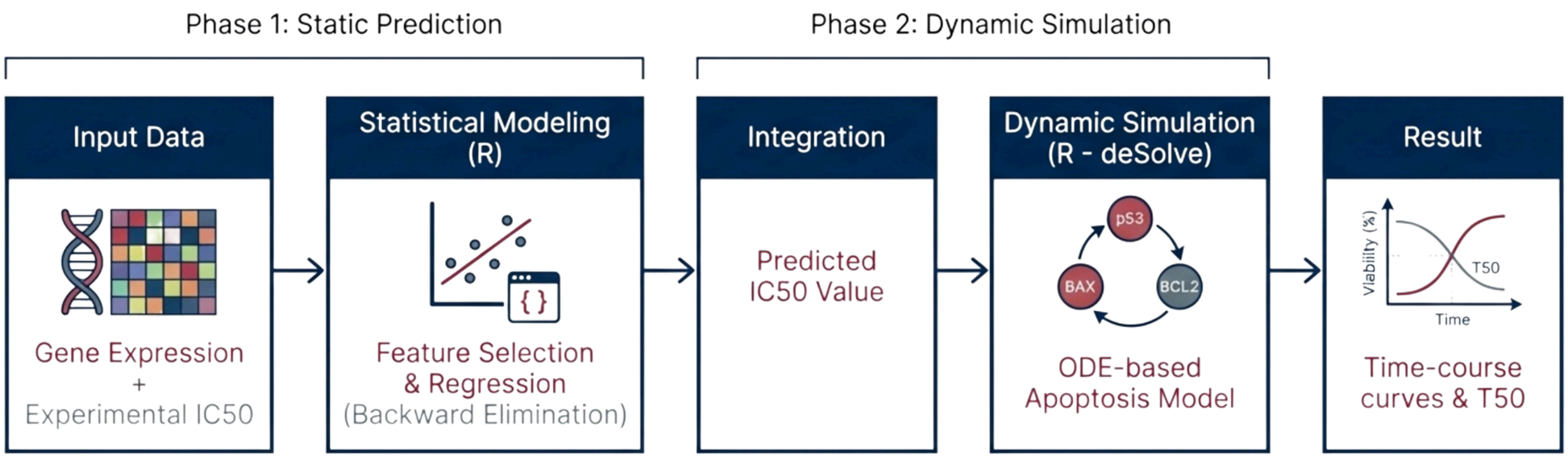
Biological Background

Doxorubicin-induced apoptosis is primarily mediated through the mitochondrial pathway. DNA damage caused by doxorubicin activates p53, a key regulator of cell fate decisions. Activated p53 promotes the expression of pro-apoptotic proteins such as BAX while suppressing anti-apoptotic proteins such as BCL-2.

The balance between BAX and BCL-2 determines mitochondrial membrane permeabilization and irreversible apoptotic commitment. Therefore, the BAX/BCL-2 ratio represents a critical molecular decision point for apoptosis.



Methodology



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References